Confidence Intervals

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MATH 240

Week 10, Lecture #1 — Ch. 12

Reminders

- **HW6** due tonight at 11:59pm
  - **HW7** will be assigned tonight, due next Monday, 11/9 at 11:59pm
- **Lab #3** due tomorrow night at 11:59pm
- Read through Chapter 12 of IMS over the next few days
- **Checkpoint #4** on Friday!
  - Open-note and done individually
  - Focused on hypothesis testing and confidence intervals
    - ★ You will not be writing R code, but you will need to interpret results from R

Objectives

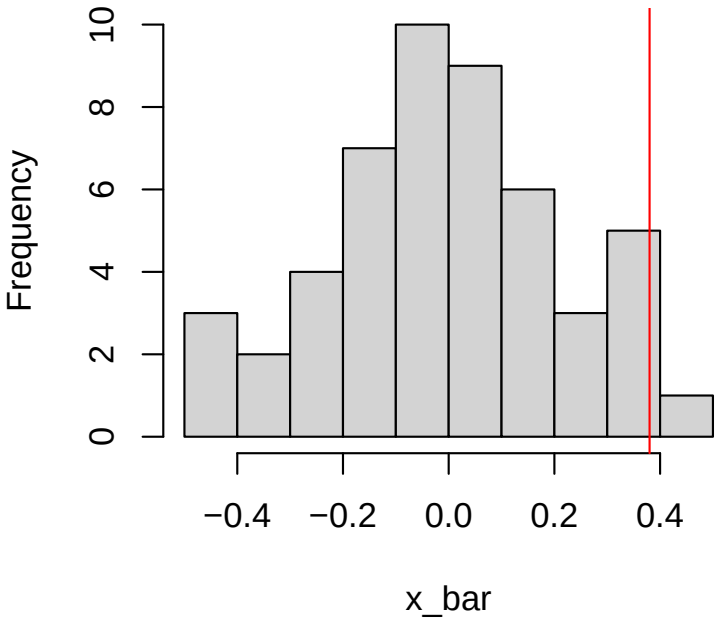
We will be exploring the use of confidence intervals (from Chapter 12 of IMS). You will:

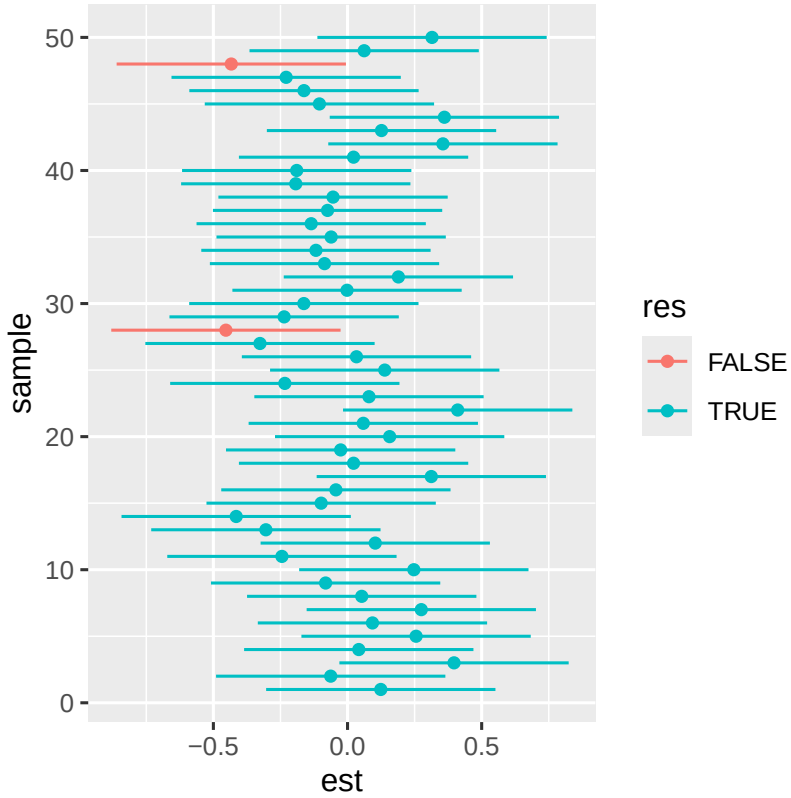
- **Define** what confidence intervals are and how we use them through computation methods (e.g. bootstrapping)
- **Run** R code to create a bootstrap random sample based on various situations
- **Identify** components of a confidence interval

From *p*-values to Confidence Intervals

- **p-value**: the probability of obtaining a test statistic as rare or more rare than our observed test statistic if the null hypothesis were true
- **Confidence interval**: plausible range of values for the population parameter
  - Centered around our test statistic

Motivation for Confidence Intervals





Motivations for Confidence Intervals

Our sample statistic is our best guess...

- However, error (i.e. variability) will exist due to the sampling process
- With this in mind, we might ask:
  1. What is the range of plausible values for the my data and statistic?
  2. Are the observed data and statistic consistent with a specific truth?

Goal: We want to capture a range of plausible values for which we may capture the true population parameter.

Example: Public Transit

Suppose we were curious about whether those in a metropolitan area were taking public transit as much as what is seen nationally. We decide to take a sample of 150 from a population of 500,000. We may get the following samples:

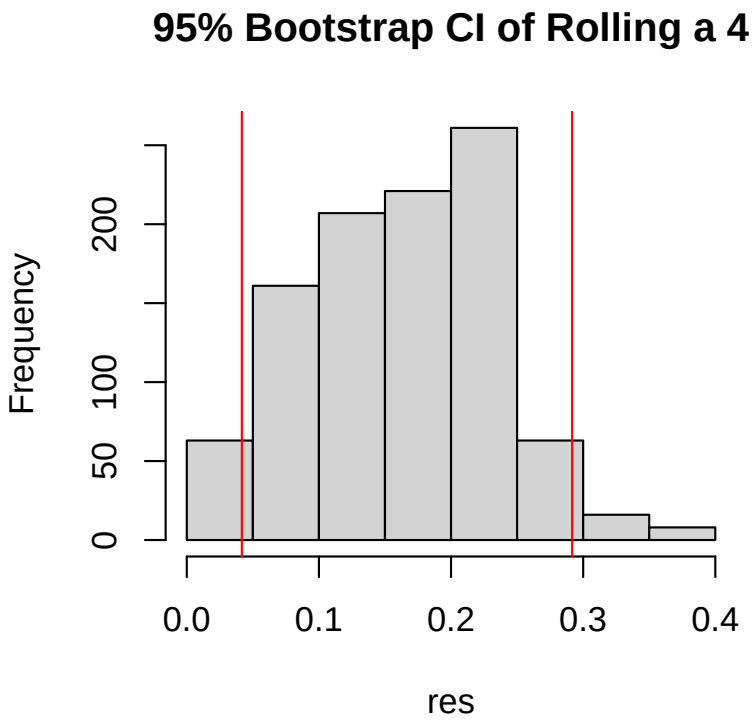
1. 4 out of 150 take public transit
2. 8 out of 150 take public transit
3. 13 out of 150 take public transit
4. 7 out of 150 take public transit
5. 9 out of 150 take public transit
6. 7 out of 150 take public transit
7. ...and so forth

Confidence Intervals

- **Confidence interval (CI)**: an interval providing a range of plausible values for the parameter, given the observed data
- **Confidence level** $(1 - \alpha)$: long-run percentage of intervals that capture the true parameter
  - Usually $1 - \alpha = 0.95$

Bootstrap Confidence Intervals

- Can look at the $100 \times (\alpha/2)$th and $100 \times (1 - \alpha/2)$th percentiles to get our **bootstrap** confidence interval
- Generating N samples where we roll 24 dice and find number of fours we rolled
- 95% bootstrap confidence interval: (0.0417, 0.2917)



What is Confidence in a Confidence Interval?

1. After N repeated samples: how many intervals contain the truth?
2. Confidence, not probability
   - Why? Confidence interval will or will not contain our null value
   - Probability comes into play when taking the sample, not so much once we have the observed data

Questions to Consider

1. Our 95% confidence interval for the probability of rolling a four was (0.0417, 0.2917). Given this, does it seem reasonable to conclude that the die wasn't fair? Why or why not?
2. What would a 100% confidence interval for the probability of rolling a four look like?
3. Assuming that they were constructed on the same sample, which would you expect to be wider: a 90% or 95% confidence interval? Do we not have enough information to tell?

Confidence Interval Code

Let's explore confidence intervals with some R code!

Activity

Use the code and change the following variables one at a time:

- `ITS`: number of iterations
- `B`: number of bootstrap samples
- `n`: number of dice rolls
- `P0`: p_0
- `ALPHA`: discernibility level

What do you notice happens to the plot (e.g. how many red lines we get) when we change these variables?